Topic - 6 ⇒ Contructor

(__init__)

A contructor is a special method in a class that automatically called when an object is created.

In python, the contructor is defined using __init__

* when you create an object, you often want intialize it with some data immediately

That is exactly what the contructor does.

Syntax:

```
class ClassName:
    def __init__(self, parameters):
        #initialize object.
```

* __init__ is a special method
* It runs automatically when object is created.
* used to assign values to instance variables.

Ex

```
class student:
    def __init__(self, name, age):
        self.name = name
        self.age = age

s1 = student("Nitheesh", 18)
print(s1.name)
print(s1.age)
```

#1 default calling

Without constructor..

- messy
- unsafe
- not scalable.

Types of constructor → Default / Parameterized

① Default:

No parameter (except self)

```
class Test:
    def __init__(self):
        print("constructor called")

t1 = Test()
```

② Parameterized

```
class Test
    def __init__(self,x):
        self.x = x
        print(x)
    def __init__(self,a,b):
        self.a = a
        self.b = b
        print(a b)
```

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t1 = Test(100)

t2 = Test(10, 20)

☆☆☆

- constructor is automatically called
- used to initialize object data.
- Defined using __init__
- First parameter must be self
- can take arguments.

☆☆ constructor doesnot create object

object is created first → then constructor initializes it

Object created

↓

constructor called

↓

value assigned.

☆ Python doesn't support constructor overloading directly. → It can be achieved by *arg / *kwarg /